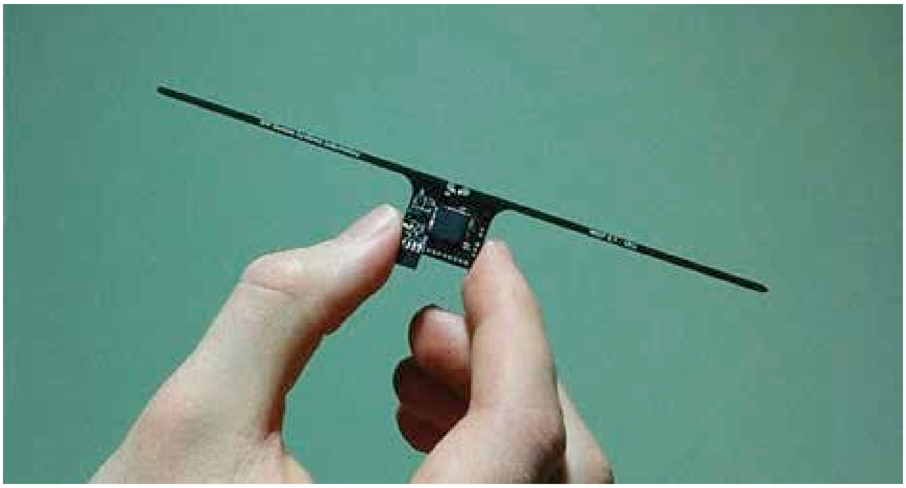
IBM Deep Thunder
IBM claims that Deep Thunder can forecast the weather to the scale of a city block **http://dgit.in/deepthunder**

The KiloCore chip
The KiloCore chip created by UC Davis boasts being the first with 1000 independant processors **http://dgit.in/kilocore**

communication or to simply just receive instructions and act upon them, now with WISP5-enabled sensors you don't need them to be connected to any power source or have them hooked into a battery module. What's more, the FRAM inside them will ensure incoming (or downstream) wireless data linkages operate at nearly negligible power draw. In fact, because the WISP5's FRAM module is so darn power efficient, the chip receiving external wireless instructions can power on instantly, download data, power off in a blink if incoming signal wavers, then power back on in a zap and continue from where it left off.

Also, a critical component of the work credited to the TU Delft part of this WISP team is when they came up with a very fast downstream data transfer mechanism dubbed Wisent on to the WISP chip. In their findings, they realized that a lot of work had gone into optimizing data upstream from an RFID tag and its corresponding client device. Curiously, however, the TU Delft found downstream data transfer from a client device to an RFID tag (similar to a WISP device) severely clunky with a lot of room for improvement. And that's exactly what they got busy with.



A chip that can doesn't need external power sounds like Sci-Fi, yet here we stand!

The research team attributes the novelty of Wisent to its ability to adaptively minimize the transfer times at varying channel conditions between the RFID tag and client device. Their experiments show that Wisent allows data transfer up to 16 times faster than a baseline, non-adaptive shortest frame case, found in most RFID implementations. Based partly on this, the team was able to successfully demonstrate how Wisent enables wireless CRFID (computational RFID) reprogramming, demonstrating the world's first wirelessly reprogrammable (software defined) CRFID.

Not just that, although diminutive and tiny, WISP-enabled chips have been implemented in complex cryptography and security scenarios also, giving it a crucial integrity layer which is essential for real-world deployments of WISP-enabled devices and solutions in the coming years.

Reality check

Promising no doubt, but we need to put things into perspective about WISP5. If you thought you'd have WISP5 toting smartphone and smart home sensors come CES 2017 or MWC 2017, time to curb some of that enthusiasm for now. There's a long road ahead before WISP comes to an off-the-shelf device which you and I can purchase.

The WISP5 module (pictured above) is supposed to be a proof of concept at the moment. It can only hold 64KB of memory and as you read earlier in the article its clock speed of 16MHz doesn't really make it a candidate for use in even modest computing devices. According to Aaron Parks, development team lead for WISP at University of Washington's Sensor Lab, "It's not going to run a video game, but it can track sensor data, do some minimal processing tasks, and communicate with the outside world."

And that's exactly where WISP's going to have an immediate impact. In the world of sensors, deeply embedded systems for extremely long-term deployments, WISP-enabled devices will make a lot of sense to a lot of companies. The breakthroughs achieved in WISP-enabled chips hold a lot of promise for the Internet of Things wave of devices.

Right now, probably a WISP-enabled sensor can't be more effective for sensing things like temperature, orientation, and acceleration, besides doing some low-intensity calculations on-the-fly inside a Fitbit-like wearable device. In time, the scale of computational prowess will only grow, and WISP-enabled devices will be more prevalent. But this is looking really far into the future.

Just think about it. In the field of healthcare, where deeply embedded sensors like pacemakers for the heart and other sensitive organs is concerned, WISP-enabled tiny, passive nodes can be inserted. Unlike existing pacemakers which need to be replaced only through invasive surgeries, these future WISP-enabled pacemakers could be reprogrammed and powered effectively through just radio waves. Similarly, in the housing and infrastructure sector, WISP-enabled sensors can be deployed in structural beams of buildings to monitor the structural integrity of a building without the need of complicated plumbing for routing wires or large-scale battery installations on site. And wouldn't it be great to have a smartphone someday that isn't constrained to the Li-ion battery it bundles with it and can work, in some small capacity, completely through wireless power harvested from radio waves? The possibilities are truly mind-boggling, if you think of it.

The researchers working hard on WISP and Wisent want to realize the dream of truly wirelessly reprogrammable software-defined battery-less computers (not just sensor nodes) whenever and wherever we want. For their sake and the rest of us digital natives, we hope that day is not too far away in the future.